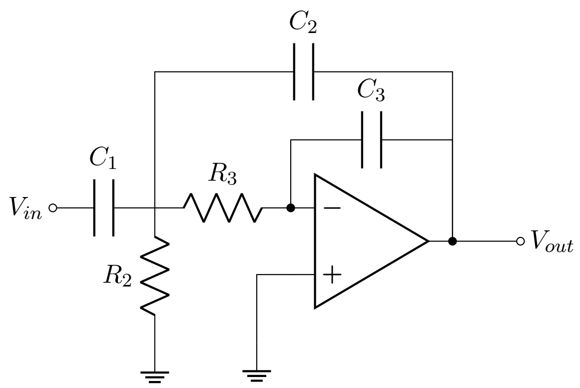
Single chanel

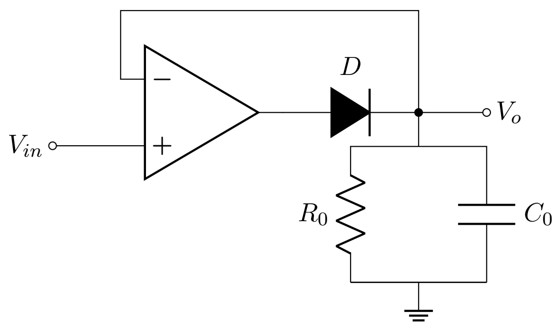
Filter.

This is a second order MFB (Multiple feed back Band Pass Filter). By changing R_2, C_1, C_2, C_3 we can change the cut-off frequency.



Peak Detector.

We need to examine how the peales change in each frequency component so that we can display them. for more reliable resault we use Op Amp based peale defector rather than single diode peale detector.



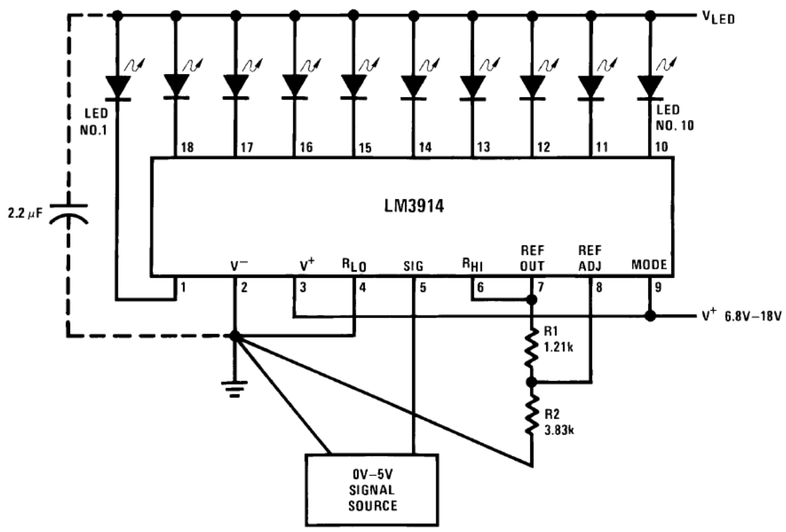
Comparator.

We need to map peale detector output to number of leds which is proportional to its amplitude. We could use Op amp compurators for this but for 10 bands which include $10 \times 10 = 100$ we require 100 opamps which is not practicle. Heance we use LM3914 IC has 10 comparators.

This Ic has two modes. Dot & Bar mode. Let's use bar mode for Visually apealling.

From Dadasheet;

TYPICAL APPLICATIONS



Ref Out V = $1.25\left(1 + \frac{R2}{R1}\right)$

$I_{LED} \cong \frac{12.5}{R1}$

$$V_{Ref} = 1.25\left(1 + \frac{R_2}{R_1}\right)$$

$$I_{LED} \cong \frac{12.5}{R_1}$$

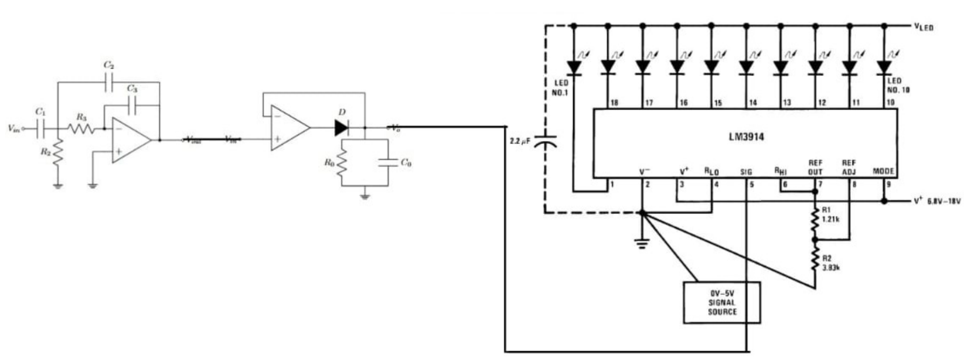
Since Audio source typical (Voud) max $\approx$ 2.5V
Choose $V_{Ref} = 2.5V = 1.25\left(1 + \frac{R_2}{R_1}\right)$
Hence $R_1 = R_2$

Choose current per led to be 10 mA

$$I_{LED} = 10 \text{ mA} = \frac{12.5}{R_1}$$

$$\Rightarrow R_1 = 1.25 \text{ k}\Omega$$

Combined Circuit per chanel.



Repeat the above circuit to create 10 chanels. Diagram of full circuit.